🚧 Depth Vision Edge Anti-Fall Detection (Edge / Cliff Detection)

1. Course Content

This course uses the **Angstrong Nuwa-HP60C 3D Depth Camera** to obtain the real-time depth distribution matrix of the ground in front of the robot. The lower one-third region of the image is taken as the ground detection region of interest (ROI). By calculating the median depth in real-time, it determines whether there are steps, table edges, or sudden ground subsidence ahead. When a sudden change in ground depth is detected (cliff or void detected), the robot automatically triggers emergency braking to prevent the chassis from falling and being damaged.

> **Important**
>
> This course is only applicable to users with the **Angstrong Nuwa-HP60C Depth Camera** version.

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2. Preparation and Running

This case requires opening 2 terminals to start the system:

2.1 Start Low-level Driver and Camera

Run in **Terminal 1** (start chassis, robotic arm and depth camera driver):

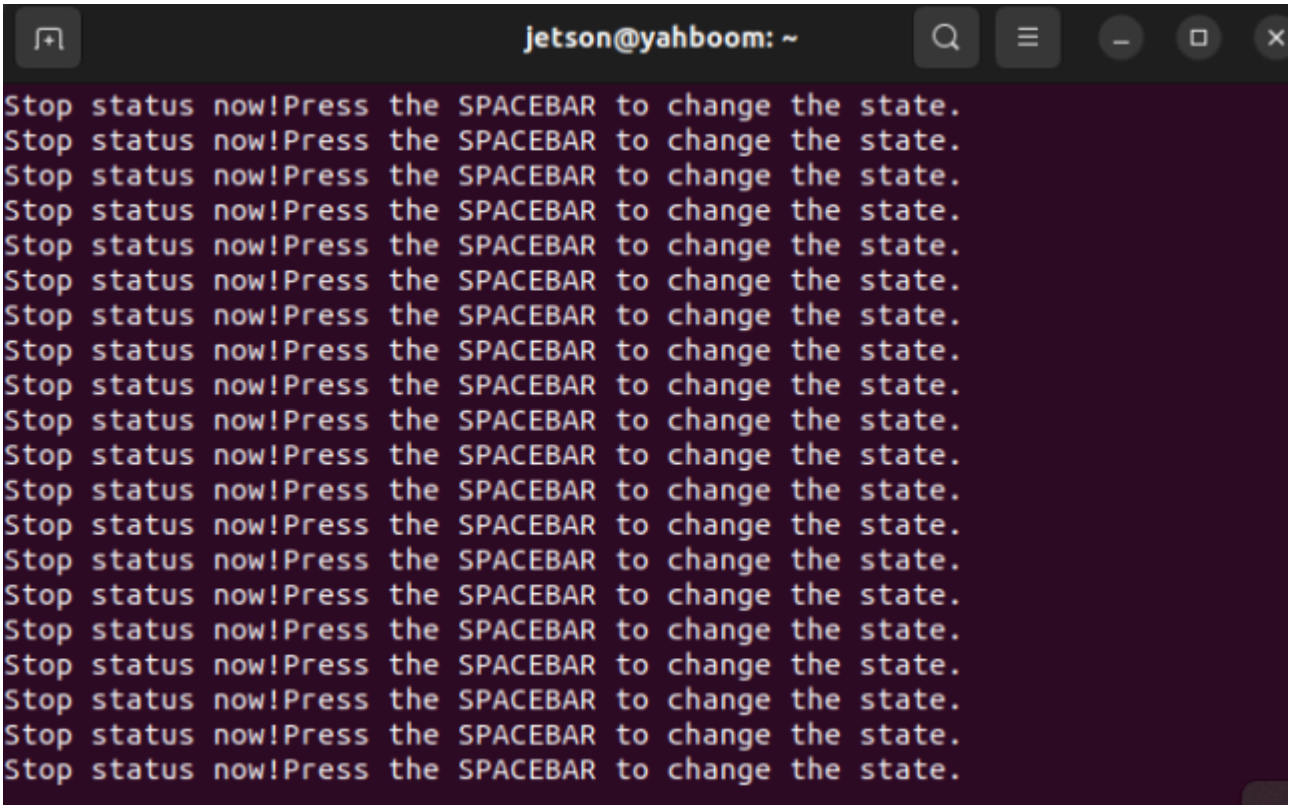
```
ros2 launch dofbot_pro_driver A1pro_camera_arm.launch.py
```

2.2 Start Edge Anti-Fall Detection Program

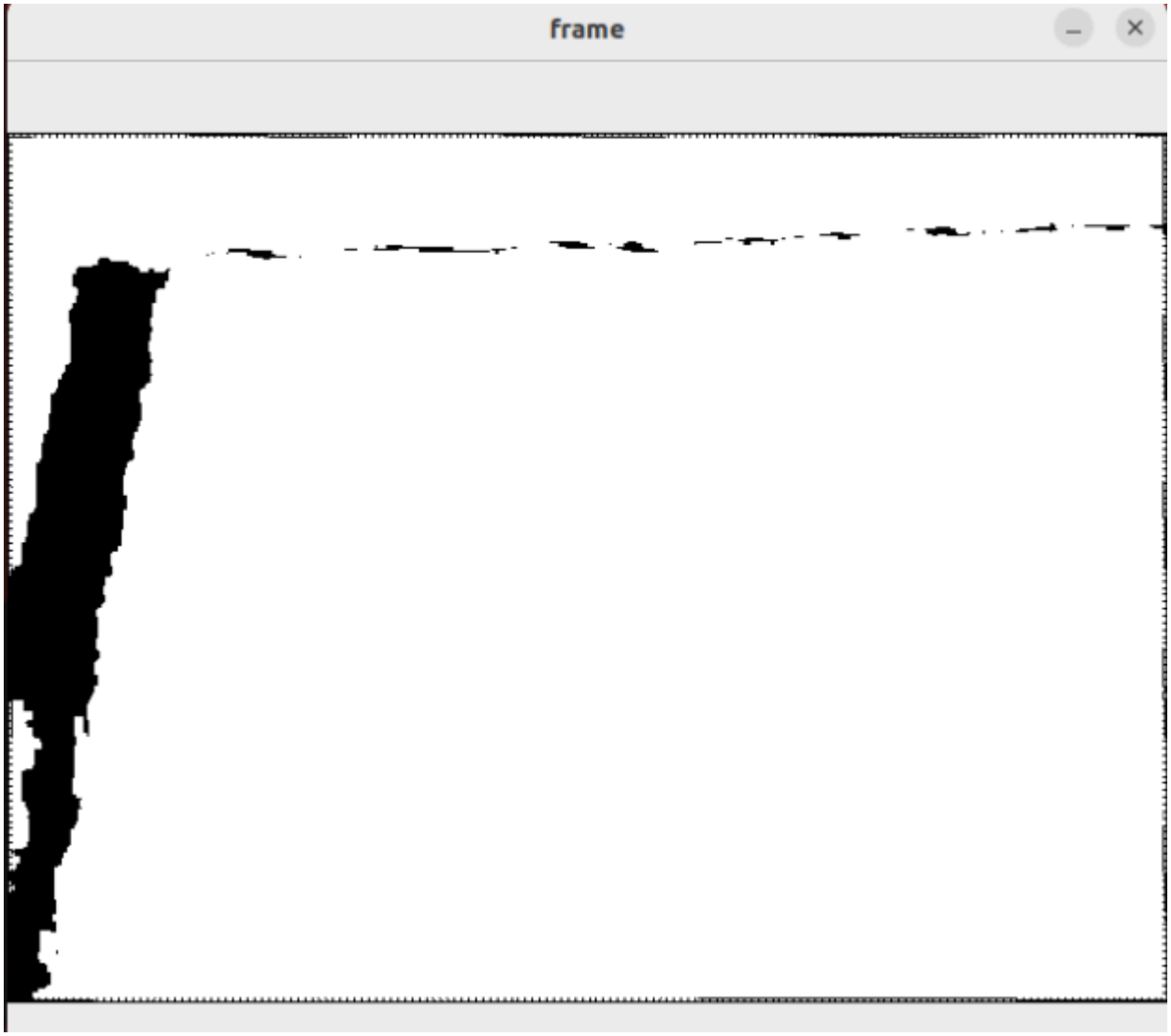
Run the edge detection node in **Terminal 2**:

```
ros2 run yahboomcar_depth Edge_Detection
```

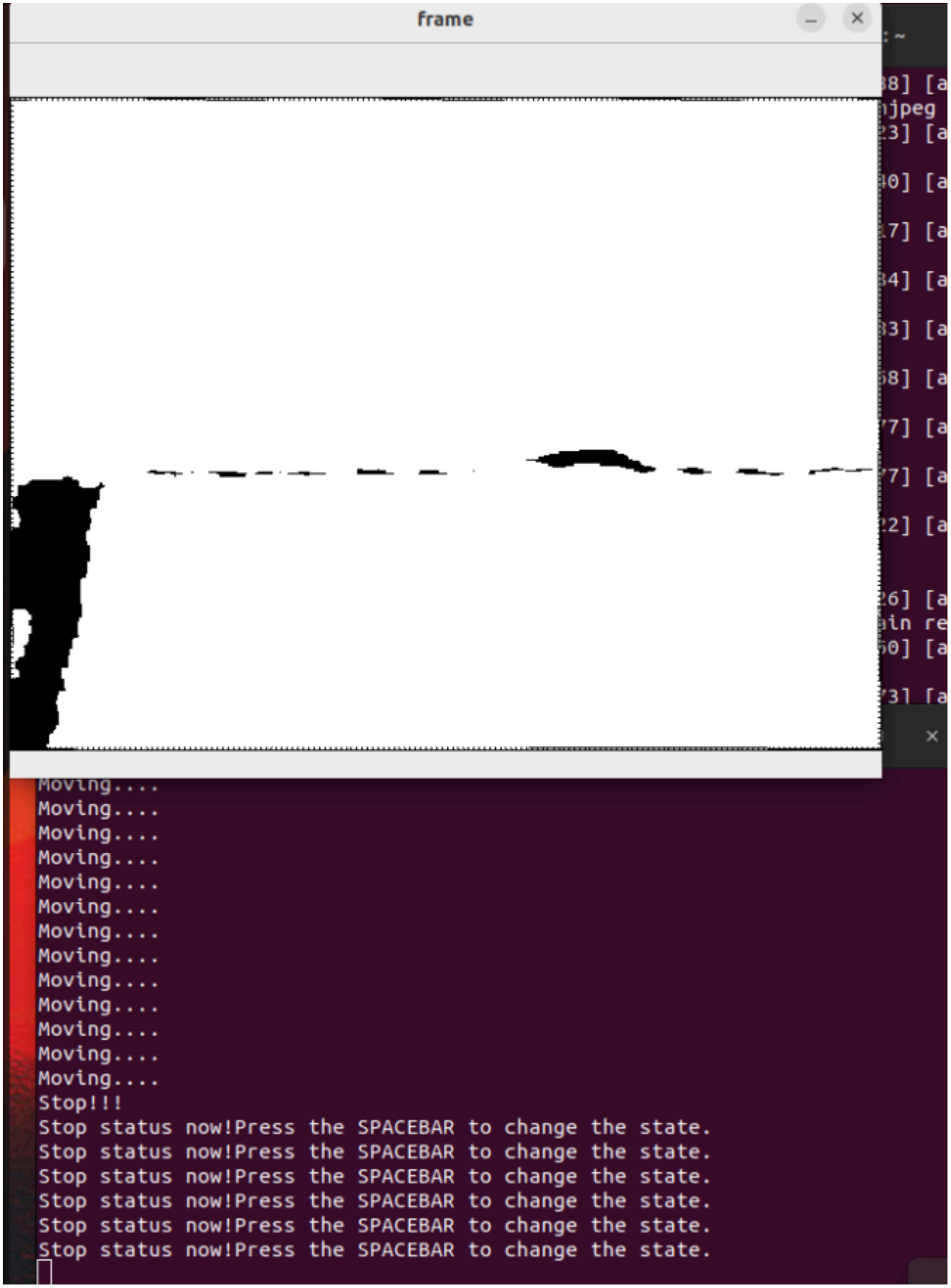
After the program starts successfully, the terminal will pop up a real-time image window:



Press the spacebar in the image window, and the robot starts moving forward smoothly:



When approaching the edge of a cliff, the robot perceives the missing ground and immediately triggers emergency automatic braking:



3. Hands-on Demo and Interaction Description

1. **Initial idle state**: After startup, the interface prompts `IDLE - Press SPACE`, and the robot is in safe standby state;
2. **Start forward cruise**: Click to focus on the OpenCV image window, press **spacebar (`Space`)**, and the robot starts forward cruising at a linear velocity of 0.1 m/s;
3. **Cliff edge emergency braking protection**: When the robot travels to a table edge or deep pit step, because the front ground is missing causing the ROI measured depth to suddenly increase (exceeding `0.4m`), the node immediately publishes a zero-speed emergency brake command to the `/cmd_vel` topic (`linear.x = 0`);
4. **Reset and retest**: If retesting is needed, move the robot back to flat safe ground, and press **spacebar** again to reactivate forward motion.

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4. Core Source Code and Algorithm Principle

Source Code Path:

```
yahboomcar_ws/src/yahboomcar_depth/yahboomcar_depth/Advanced/Edge_Detection.py
```

4.1 Dynamic Ground ROI Extraction and Median Depth Filtering

To avoid interference from distant walls or obstacles in the upper field of view on ground distance measurement, the algorithm crops only the lower $\frac{1}{3}$ region of the depth image and uses median filtering (`np.median`) to effectively remove noise caused by specular reflections or zero-depth blind spots:

```python
# Edge_Detection.py —— Ground ROI extraction and filtering
def get_DepthImageCallBack(self, msg):
    depth_image = self.depth_bridge.imgmsg_to_cv2(msg, '32FC1')
    h, w = depth_image.shape[:2]
    # Crop the lower 1/3 region of the image as ground detection ROI
    bottom_third_start = h * 2 // 3
    roi = depth_image[bottom_third_start:, :].astype(np.float32)

    # Filter invalid zero-depth blind spots, calculate median depth of valid pixels
    valid_pixels = roi[roi > 0]
    median_depth = (np.median(valid_pixels) / 1000.0) if len(valid_pixels) > 0 else 0.0
```

4.2 Cliff State Machine and Emergency Braking Control Loop

Compare the calculated current physical distance with the safety threshold `depth_threshold` (0.4m):

```
# Edge_Detection.py -- Velocity publishing and cliff braking control
if self.move_flag:
    if median_depth > self.depth_threshold:
        # Front depth sudden change exceeds threshold, judge as cliff edge, immediately
publish emergency brake
        self.pubVel(0, 0, 0)
        self.move_flag = False
        self.get_logger().warn(
            f"[CLIFF DETECTED] Depth={median_depth:.3f}m > {self.depth_threshold}m,
STOPPING!")
    else:
        # Ground is flat and normal, maintain set linear velocity for uniform forward motion
        self.pubVel(self.lin_x, 0, 0)
else:
    self.pubVel(0, 0, 0)
```